

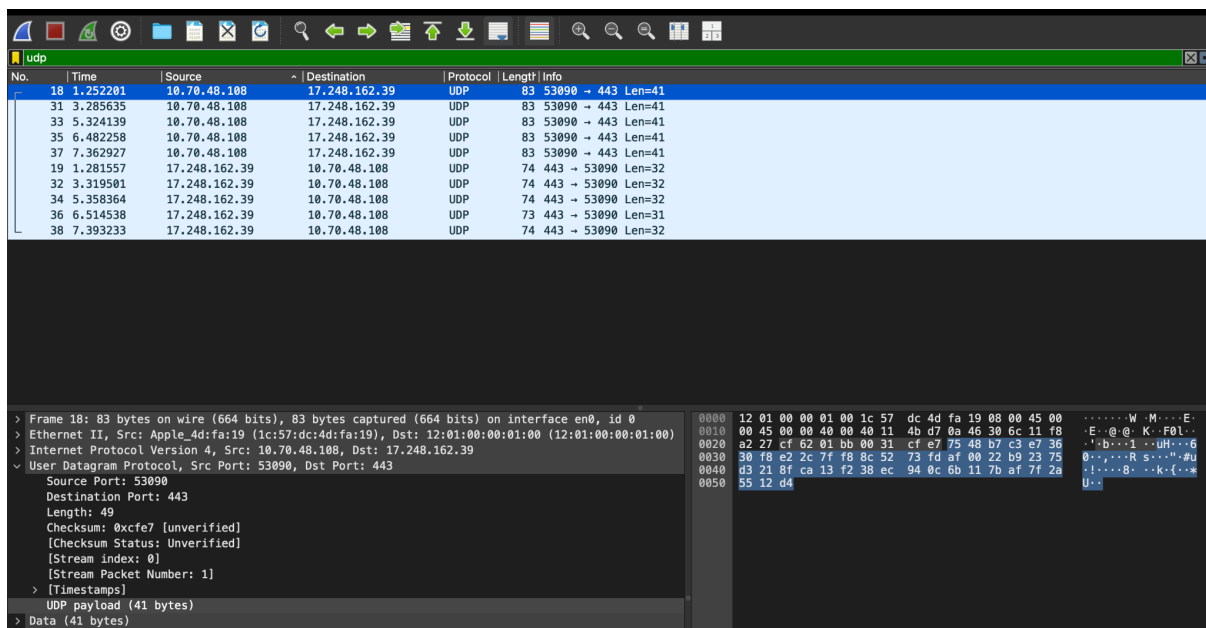
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CN Assignment-5

1. There are 4 fields in the UDP header:

- Source Port
- Destination Port
- Length
- Checksum



2. Each of the header fields is of 2 bytes.

3. The value Length field is the length of 49 bytes.

Verification: Length = Size of UDP Payload + Size of UDP Header

Here, Size of UDP Payload = 41 bytes

Size of UDP Header = 8 bytes = 4 headers * 2 bytes

Length = 41 + 8 = 49 bytes.

4. The length of the UDP field is 2 bytes (= 16 bits). This shows that the maximum number of bytes which can be included in a UDP is 65535 (= $2^{16} - 1$).

Subtracting the UDP header length i.e. 8 bytes,
Maximum bytes of UDP payload = $65535 - 8 = 65527$ bytes.

5. UDP Header is of 16 bits. Since counting starts from 0, the largest possible source port number is 65535 ($= 2^{16} - 1$).
6. Protocol Number for UDP: 17 (in decimal) or 0x11 (in hexadecimal).
7. Observations:
 - The source port of the 1st packet becomes the destination port for the 2nd packet.
 - The destination port of 1st packet is the source port for the 2nd packet.

Wireshark packet capture showing packet 18. The packet list shows a UDP packet from 10.70.48.108 to 17.248.162.39, port 53090 to 443. The packet details show Ethernet II, Internet Protocol Version 4, and User Datagram Protocol. The packet bytes pane shows the raw data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
18	1.252201	10.70.48.108	17.248.162.39	UDP	83	53090 → 443 Len=41

Frame 18: 83 bytes on wire (664 bits), 83 bytes captured (664 bits) on interface en0, id 0
> Ethernet II, Src: Apple_4d:fa:19 (1c:57:dc:4d:fa:19), Dst: 12:01:00:00:01:00 (12:01:00:00:01:00)
> Internet Protocol Version 4, Src: 10.70.48.108, Dst: 17.248.162.39
> User Datagram Protocol, Src Port: 53090, Dst Port: 443
Source Port: 53090
Destination Port: 443
Length: 49
Checksum: 0xcfe7 [unverified]
[Checksum Status: Unverified]
[Stream index: 0]
[Stream Packet Number: 1]
> [Timestamps]
UDP payload (41 bytes)
> Data (41 bytes)

Wireshark packet capture showing packet 19. The packet list shows a UDP packet from 17.248.162.39 to 10.70.48.108, port 443 to 53090. The packet details show Ethernet II, Internet Protocol Version 4, and User Datagram Protocol. The packet bytes pane shows the raw data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
19	1.281557	17.248.162.39	10.70.48.108	UDP	74	443 → 53090 Len=32

Frame 19: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface en0, id 0
> Ethernet II, Src: Hewlett-Packard_4d:00:00:00:00:00 (e4:de:40:ac:4d:00), Dst: Apple_4d:fa:19 (1c:57:dc:4d:fa:19)
> Internet Protocol Version 4, Src: 17.248.162.39, Dst: 10.70.48.108
> User Datagram Protocol, Src Port: 443, Dst Port: 53090
Source Port: 443
Destination Port: 53090
Length: 40
Checksum: 0xe9c1 [unverified]
[Checksum Status: Unverified]
[Stream index: 0]
[Stream Packet Number: 2]
> [Timestamps]
UDP payload (32 bytes)
> Data (32 bytes)